

LIVE

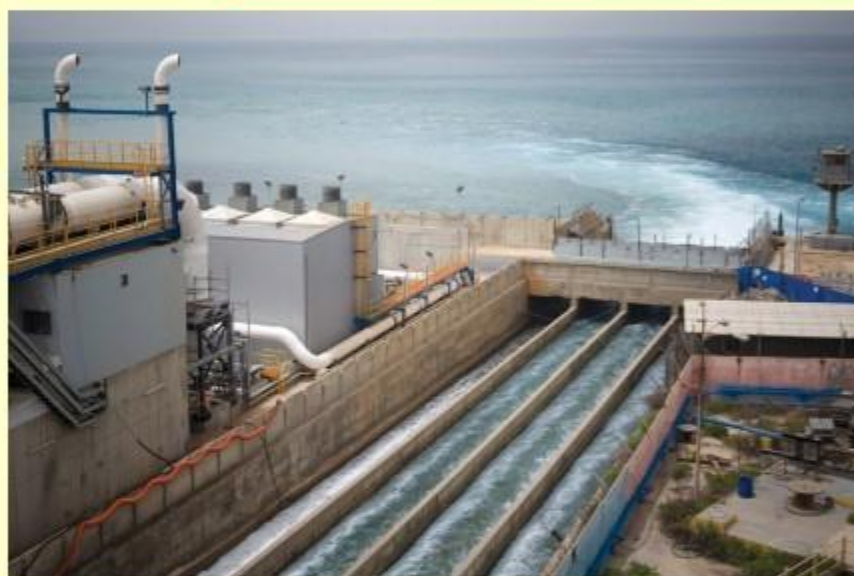


NEWSPAPER

4th July, 2017

ITEM WISE EDITORIALS

27th WEEK



WATER & ISRAEL

UPSC CSE – GS III

GS III ... DIFFERENT TYPES OF IRRIGATION AND IRRIGATION SYSTEMS, STORAGE.

4

Irrigation tips from an arid country

Prime Minister Modi's visit to Israel should build on the partnership between the two countries in water management



ASHOK GULATI AND
GAYATHRI MOHAN

WHEN PRIME MINISTER Narendra Modi visits Israel this week, he will become the first Indian PM to do so. The Indo-Israel partnership is likely to extend to new areas. Israel is already an important supplier of defence equipment to India; the cooperation is likely to extend to water and agriculture.

In the run-up to the PM's visit, ICRIER and the Israeli Embassy in India collaborated on a brainstorming session on a potential partnership between the two countries on water. The meet was attended by several experts from India and Israel. A few takeaways from this meet could result in MoUs between the two countries.

It is well-known that India is water stressed. The country's annual per capita availability of water is less than 1,500 cubic meters (cum) and that scenario is likely to worsen. Currently, about 78 per cent of fresh water in India is used for agriculture. But Israel has per capita water availability of less than 200 cum. It is an extremely water scarce region. Yet, the country exports high-value agri-produce to Europe and many other parts of the world. If India has to learn about augmenting water supplies and using water efficiently, there is nothing better than learning from Israel's experience. The country is known as the land of innovations in water management, be it drip irrigation, recycling of urban wastewater for use in agriculture or desalination of seawater for drinking purposes.

The Pradhan Mantri Krishi Sinchayee Yojana (PMKSY) encapsulates the PM's vision of "Kisat Khet ke paani" and "more crop per drop". Micro-irrigation technologies — drip irrigation and sprinklers — are not alien to India. In fact, Jain Irrigation Systems Limited — a multinational outfit based in Jalgaon, Maharashtra — has done well in this area, and has acquired the highest stakes in the second largest Israeli drip irrigation company, Netafim. At Jalgaon, the company is innovating on this technology with the simple motto of "Small flows, big revolutions". India has about nine million hectares under micro-irrigation. It can scale this up several

shown much appetite to reform the regime of high subsidies on water and power, which is leading to their inefficient use.

As per Israel's water law, all water in the country is common property resource. The government does accounting for every drop of water, ensuring good water governance. As a result, in 2013, of the total water available in Israel almost 16 per cent came from desalinating seawater and another 22 per cent came from recycling wastewater. Interestingly, almost 62 per cent of the irrigation water used in Israel comes from recycled and brackish water, thus saving potable water for domestic use. Such recycling endeavours — of course with proper treatment that make wastewater safe — has relevance in India. Can the Modi government make recycling of wastewater for use in peri-urban agriculture an integral part of its smart city initiative? India does have the technology for this purpose. In fact, the municipality-owned sewage treatment plants in Surat and Nagpur supply treated wastewater to textile factories and thermal power stations, respectively.

Desalination technology can benefit India which has a long coastline of about 7,500 km. The Nemmeli desalination plant in Tamil Nadu, a source of potable water for people in Chennai, and the Jamnagar desalination plant in Gujarat are some of the joint Indo-Israeli initiatives in desalinating sea waters. Adoption of desalination technologies is at a nascent stage in India, owing to the high capital cost involved. The key issue for India is who bears the cost of such technologies — the government and/or the beneficiaries? This is an issue of public policy that needs to be sorted out.

Another area of potential cooperation is cleaning polluted rivers. As per government reports, between 2009 and 2015, the number of polluted rivers and river stretches in India has almost doubled. Almost 63 per cent of the total sewage generated in the country is discharged into water bodies without any treatment resulting in acute river pollution. The Ganga and Yamuna, considered holy, are ironically the two most polluted rivers. Programmes to clean up the Ganges have, over 35 years, failed to achieve significant success. Recently, the Delhi Government has roped in the Israeli firm, Ayala Water and Ecology Limited, to clean up an eight km stretch of the Yamuna. Depending upon the success of this venture, the country can scale up the role of Israeli firms in cleaning up other rivers.

An integrated approach to solving India's emerging water crisis, involving government to government, governance to busi-



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Date : 3rd July, 2017



Cyprus

Mediterranean Sea

ISRAEL

SYRIA

LEBANON

IRAQ

JORDAN

EGYPT

Gulf of Aqaba

SAUDI ARABIA

ISRAEL & WATER CONSERVATION



- 1** *In Israel, the per capita water availability is less than 200 cubic meters and where as for our country, it is around 1,500 cubic meters*
- 2** *Israel is an extremely water scarce region.*
- 3** *Still the country exports high value agri-produce to Europe and other parts of the world.*
- 4** *The country is pioneer in the water management, whether it is drip irrigation, recycling of urban waste water for use in agriculture or desalination of seawater for drinking purposes.*



LOOK AT THE SCENARIO IN INDIA

PER CAPITA AVAILABILITY

STIPULATIONS

WATER STRESS
CONDITION ... 1,700 m³
WATER SCARCITY
CONDITION ... 1,000 m³

AVAILABILITY

2017 ... 1,545 m³
2025 (expected) ... 1,341 m³
2050 (expected) ... 1,140 m³

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It is well-known that India is water stressed. The country's annual per capita availability of water is less than 1,500 cubic meters (cum) and that scenario is likely to worsen. Currently, about 78 percent of fresh water in India is used for agriculture. But Israel has per capita water availability of less than 200 cum. It is an extremely water scarce region. Yet, the country exports high-value agri-produce to Europe and many other

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WHAT ABOUT INDIA?

- 1 India is water stressed, but the country's per capita availability of water is less than 1,500 cubic meters.
- 2 About 78% of fresh water in India is used for agriculture.
- 3 PMKSY now talks about 'Har Khet ko Pani' and 'More crop per drop'.
- 4 India has made some baby steps under micro-irrigation.
- 5 At present, about 9 million hectares is around micro irrigation and India can scale up several times.



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DRIP IRRIGATION



SPRINKLER IRRIGATION



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WHY SO SPECIAL ABOUT ISRAEL WATER LAW?

- 1 All water in the country is common property resource.
- 2 The Govt. accounts for every drop of water.
- 3 16% of total water availability came from desalinating seawater, 22% came from recycling waste water.
- 4 Almost 62% of irrigation water comes from recycled and brackish water.
- 5 It saves potable water for domestic use.



The Indian **EXPRESS**

Date : 3rd July, 2017



WHAT CAN BE THE LESSONS INDIA LEARN FROM ISRAEL IN WATER MANAGEMENT?



- 1** *India has to go in a big way for micro irrigation i.e. drip and sprinkler irrigation systems.*
- 2** *Right policy environment with mechanism of pricing water and power for irrigation purposes is to be introduced.*
- 3** *Recycling of waste water is the need of the hour and it can be an integral part of Smart Cities initiative. At present, Municipality owned sewage treatment plants in Surat and Nagpur supply treated waste water to textile factories and thermal power stations respectively.*
- 4** *Desalination technology can benefit India in a big way as India has got a coastline of about 7,500 kms. The Nemmeli Desalination Plant in Tamil Nadu and the Jamnagar Desalination Plant in Gujarat are some of the joint Indo-Israeli initiatives in desalinating seawaters. Govt. has to take a policy decision with regard to the capital costs involved.*
- 5** *Another area of cooperation can be in cleaning the polluted rivers. As per the Govt. reports, between 2009 and 2015, the number of polluted rivers and river stretches in India has almost doubled.*



WHAT CAN BE THE LESSONS INDIA LEARN FROM ISRAEL IN WATER MANAGEMENT?



6 62% of total sewage generated in the country is discharged into water bodies without any treatment resulting in acute river pollution.



7 Ganga and Yamuna are the most polluted rivers in the country. Recently, Delhi Govt. has roped in the Israeli firm, Ayala Water and Ecology Ltd. to clean up an 8 km stretch of Yamuna. Depending on its success, it can be extended to other reaches of rivers.



An integrated approach in solving India's emerging water crisis involving Govt-to-Govt, Govt-to-Business and Business-to-Business interactions between Indians and Israeli agencies has much to offer not only between the two countries but also to several countries in African region, which are starved of water.



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